Cambridge O Level Computer Science

Chapter 2: Data Transmission

Lecture notes with Check-Yourself questions and a full answer key.

In this chapter you will learn about:

- Types and methods of data transmission: data packets (header, payload, trailer), packet switching and the role of routers, serial vs parallel transmission, simplex / half-duplex / full-duplex, and the Universal Serial Bus (USB).
- Methods of error detection: why error checking is needed; parity checks, checksums and echo checks after transmission; check digits for data entry errors; automatic repeat requests (ARQs).
- Encryption: why it is needed; symmetric and asymmetric encryption; public and private keys.

Data is constantly moved from one device to another, whether the devices are in the same building or thousands of kilometres apart. Whenever data is transmitted, three things matter: how the data is sent, how errors can be detected (and whether the data can be recovered), and how encryption stops intercepted data from being useful to the wrong person.

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2.1 Types and methods of data transmission

2.1.1 Data packets

Data sent over long distances is broken up into small data packets (sometimes called datagrams), typically about 64 KiB each. Small packets are much easier to control than one long continuous stream of data. The main advantage: each packet can be sent along a different route to its destination, which is very useful if one route is broken or very busy. The only real drawback is that the packets must be reassembled into the correct order when they arrive.

Packet structure

Every packet has three parts: a header, the payload, and a trailer.

- Header: the IP address of the sending device, the IP address of the receiving device, the sequence number of the packet (so packets can be reassembled in the right order), and the packet size (so the receiver can check that every packet

arrived intact). The header often also states how many packets there are in total.
- Payload: the actual data being carried, usually about 64 KiB.
- Trailer: a marker identifying the end of the packet (so packets can be separated from each other), plus an error-checking value.

The error check in the trailer is a cyclic redundancy check (CRC). It works like this:

1. Before sending, the sending computer counts all the 1-bits in the payload and stores the total as a hex value in the trailer.
2. When the packet arrives, the receiving computer recounts the 1-bits in the payload.
3. It compares its total with the value in the trailer. If they match, the packet arrived error-free; if not, the packet must be re-sent.

Worked example: A packet payload contains 11010100 01111000. Show the CRC value stored in the trailer.

Count the 1-bits: 11010100 has four 1-bits; 01111000 has four 1-bits.

Total = 4 + 4 = 8.

8 in hex is 8, so the trailer stores the value 8.

At the destination the receiver recounts: if it also gets 8, the payload is error-free.

Check yourself

4. 1. Explain why data sent over long distances is broken into packets instead of being sent as one continuous stream.
5. 2. List the four items found in a packet header and explain the purpose of the sequence number.
6. 3. Describe how a cyclic redundancy check (CRC) detects a transmission error.
7. 4. A payload contains 10110011 11000011. The trailer stores the hex value C. The receiver counts eleven 1-bits. Has an error occurred? Explain.

Packet switching

Packet switching is a method of data transmission in which a message is broken into packets and each packet is sent independently from sender to receiver. At the

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destination the packets are reassembled into the correct order using the sequence numbers in their headers.

Between the sender and receiver there are nodes, and each node contains a router. A router receives a packet, looks at the destination IP address in its header, and decides which route the packet should take next. Routing depends on how busy each node is: the shortest available path is always chosen, but that may not be the shortest possible path, because parts of the network can be too busy or unsuitable. As a result, packets often arrive in a different order from the order in which they were sent.

Benefits of packet switching:

- No single communication line is tied up for the whole message.
- Failed, busy or faulty lines are overcome by simply re-routing packets.
- It is relatively easy to expand usage of the network.
- A high data transmission rate is possible.

Drawbacks of packet switching:

- Packets can be lost and need to be re-sent.
- It is more prone to errors with real-time streaming (for example, a live sports event over the internet).
- There is a delay at the destination while packets are reordered.

Note on lost packets: packets can keep bouncing from router to router and never reach their destination, which would eventually clog the network. To prevent this, a hop number is added to each packet header and reduced by 1 every time the packet leaves a router. When it reaches zero without the packet arriving, the packet is deleted, the receiver flags it as missing, and a re-send is requested. (Hopping is not on the syllabus; it is included here for completeness.)

Check yourself

8. 1. Describe the role of a router during packet switching.
9. 2. Explain why packets sent from the same message can arrive at the destination in a different order.
10. 3. Give two benefits and two drawbacks of packet switching.
11. 4. Explain why packet switching can cause problems for a live video stream.

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2.1.2 Methods of data transmission

When transmitting data, three factors are considered (usually handled by a communication protocol): the direction of transmission, the method of transmission (how many bits go at once), and synchronisation (keeping received data in the correct order).

Simplex, half-duplex and full-duplex (direction of transmission)

- Simplex: data travels in ONE DIRECTION ONLY, for example from a computer to a printer.
- Half-duplex: data travels in BOTH DIRECTIONS but NOT AT THE SAME TIME, for example a walkie-talkie (you can send or receive, but not both at once).
- Full-duplex: data travels in BOTH DIRECTIONS AT THE SAME TIME, for example a broadband internet connection.

Serial and parallel (method of transmission)

- Serial data transmission: data is sent ONE BIT AT A TIME over a SINGLE wire or channel, as a single stream (for example 1 0 0 1 1 0 1 1). Serial can be simplex, half-duplex or full-duplex.
- Parallel data transmission: SEVERAL BITS (usually one byte) are sent AT THE SAME TIME over several wires or channels. Each wire carries one bit. Parallel can also be simplex, half-duplex or full-duplex.

Serial vs parallel comparison:

Serial	**Parallel**
Slower transmission rate than parallel	Faster transmission rate than serial
Less risk of external interference (fewer wires)	More risk of interference
More reliable over long distances	Works well over short distances (for example, internal pathways on circuit boards)
Bits cannot arrive out of order (no skewing)	Bits can arrive out of synchronisation (skewing); worse over longer distances and longer wires
Cheaper: less hardware needed	More expensive: more hardware needed
Used for long distances and small amounts of data (for example, USB, telephone lines)	Used when speed matters and distance is short; internal computer circuits use parallel

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